



## GSFA Online

Updated up to the 48<sup>th</sup> Session of the Codex Alimentarius Commission (2025)

### FOOD ADDITIVE DETAILS

#### Calcium alginate (404)

##### Functional Classes

- Antifoaming agent
- Bulking agent
- Carrier
- Foaming agent
- Gelling agent
- Glazing agent
- Humectant
- Sequestrant
- Stabilizer
- Thickener

 [Click here to search the FAO JECFA database for the specifications of additive\(s\) with INS No. 404](#)

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CSV (Tables)

#### GSFA Provisions for Calcium alginate

| Number                                                                                      | Food Category                                                                                           | Max Level | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  06.4.2   | Dried pastas and noodles and like products                                                              | GMP       |  Note 256                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  01.2.1.2 | Fermented milks (plain), heat-treated after fermentation                                                | GMP       |  Note 234                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  01.2.1.1 | Fermented milks (plain), not heat-treated after fermentation                                            | GMP       |  Note 634<br> Note 235<br> Note 234                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|  09.2.2   | Frozen battered fish, fish fillets, and fish products, including mollusks, crustaceans, and echinoderms | GMP       |  Note 63                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|  10.2.2   | Frozen egg products                                                                                     | GMP       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  09.2.1   | Frozen fish, fish fillets, and fish products, including mollusks, crustaceans, and echinoderms          | GMP       |  Note XS315<br> Note XS165<br> Note XS36<br> Note XS191<br> Note XS312<br> Note XS92<br> Note XS292<br> Note XS190<br> Note XS95 |
|  10.2.1   | Liquid egg products                                                                                     | GMP       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  11.4     | Other sugars and syrups (e.g. xylose, maple syrup, sugar toppings)                                      | GMP       |  Note 258                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  01.4.1   | Pasteurized cream (plain)                                                                               | GMP       |  Note XS288                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|  01.4.2   | Sterilized and UHT creams, whipping and whipped creams, and reduced fat creams (plain)                  | GMP       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  04.1.1.2 | Surface-treated fresh fruit                                                                             | GMP       |  Note 454                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

Note: Unless otherwise specified, food additive provisions apply to the food category indicated (e.g. Dairy), as well as to all subcategories of that category (e.g. Cheese, Ripened Cheese, etc.).

## GSFA Table 3 Provisions

**Calcium alginate** is a food additive that is included in **Table 3**, and as such may be used in the following foods under the conditions of good manufacturing practices (GMP) as outlined in the Preamble of the Codex GSFA. Although not listed below, Calcium alginate could also be used in heat-treated butter milk of food category 01.1.1 and spices of food category 12.2.1. Note that food categories listed in the **Annex to Table 3** were excluded accordingly. **Calcium alginate** is acceptable in foods conforming to the following commodity standards: CS 70-1981 (for use in packing media only), CS 94-1981 (for use in packing media only), CS 119-1981 (for use in packing media only), CS 221-2001, CS 273-1968, CS 275-1973, CS 253-2006 (see functional class table and footnote).

**Calcium alginate** is a: Antifoaming agent, Humectant, Stabilizer, Thickener

Any Antifoaming agent listed in Table 3 can be acceptable for use in all products conforming to CS 117-1981, CS 256-2007.

Any Humectant listed in Table 3 can be acceptable for use in all products conforming to CS 117-1981.

Any Stabilizer listed in Table 3 can be acceptable for use in all products conforming to CS 117-1981, CS 256-2007.

Any Thickener listed in Table 3 can be acceptable for use in all products conforming to CS 117-1981, CS 256-2007, CS 306-2011.

| Number                                                                                             | Food Category                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>01.1.4</b>     | <b>Flavoured fluid milk drinks</b>                                                                                                                                                                                                                 |
|  <b>01.3</b>       | <b>Condensed milk and analogues (plain)</b>                                                                                                                                                                                                        |
|  <b>01.4.3</b>     | <b>Clotted cream (plain)</b>                                                                                                                                                                                                                       |
|  <b>01.4.4</b>     | <b>Cream analogues</b>                                                                                                                                                                                                                             |
|  <b>01.5</b>       | <b>Milk powder and cream powder and powder analogues (plain)</b>                                                                                                                                                                                   |
|  <b>01.6.1</b>     | <b>Unripened cheese</b>                                                                                                                                                                                                                            |
|  <b>01.6.2</b>     | <b>Ripened cheese</b>                                                                                                                                                                                                                              |
|  <b>01.6.4</b>     | <b>Processed cheese</b>                                                                                                                                                                                                                            |
|  <b>01.6.5</b>     | <b>Cheese analogues</b>                                                                                                                                                                                                                            |
|  <b>01.7</b>       | <b>Dairy-based desserts (e.g. pudding, fruit or flavoured yoghurt)</b>                                                                                                                                                                             |
|  <b>01.8.1</b>     | <b>Liquid whey and whey products, excluding whey cheeses</b>                                                                                                                                                                                       |
|  <b>02.2.2</b>     | <b>Fat spreads, dairy fat spreads and blended spreads</b>                                                                                                                                                                                          |
|  <b>02.3</b>      | <b>Fat emulsions mainly of type oil-in-water, including mixed and/or flavoured products based on fat emulsions</b>                                                                                                                                 |
|  <b>02.4</b>     | <b>Fat-based desserts excluding dairy-based dessert products of food category 01.7</b>                                                                                                                                                             |
|  <b>03.0</b>     | <b>Edible ices, including sherbet and sorbet</b>                                                                                                                                                                                                   |
|  <b>04.1.2</b>   | <b>Processed fruit</b>                                                                                                                                                                                                                             |
|  <b>04.2.2.2</b> | <b>Dried vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweeds, and nuts and seeds</b>                                                                                                         |
|  <b>04.2.2.3</b> | <b>Vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), and seaweeds in vinegar, oil, brine, or soybean sauce</b>                                                                                      |
|  <b>04.2.2.4</b> | <b>Canned or bottled (pasteurized) or retort pouch vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), and seaweeds</b>                                                                               |
|  <b>04.2.2.5</b> | <b>Vegetable (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweed, and nut and seed purees and spreads (e.g., peanut butter)</b>                                                                          |
|  <b>04.2.2.6</b> | <b>Vegetable (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweed, and nut and seed pulps and preparations (e.g. vegetable desserts and sauces, candied vegetables) other than food category 04.2.2.5</b> |
|  <b>04.2.2.8</b> | <b>Cooked or fried vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), and seaweeds</b>                                                                                                               |
|  <b>05.0</b>     | <b>Confectionery</b>                                                                                                                                                                                                                               |
|  <b>06.3</b>     | <b>Breakfast cereals, including rolled oats</b>                                                                                                                                                                                                    |
|  <b>06.4.3</b>   | <b>Pre-cooked pastas and noodles and like products</b>                                                                                                                                                                                             |
|  <b>06.5</b>     | <b>Cereal and starch based desserts (e.g. rice pudding, tapioca pudding)</b>                                                                                                                                                                       |
|  <b>06.6</b>     | <b>Batters (e.g. for breading or batters for fish or poultry)</b>                                                                                                                                                                                  |
|  <b>06.7</b>     | <b>Pre-cooked or processed rice products, including rice cakes (Oriental type only)</b>                                                                                                                                                            |
|  <b>06.8</b>     | <b>Soybean products (excluding soybean-based seasonings and condiments of food category 12.9)</b>                                                                                                                                                  |
|  <b>07.0</b>     | <b>Bakery wares</b>                                                                                                                                                                                                                                |
|  <b>08.2</b>     | <b>Processed meat, poultry, and game products in whole pieces or cuts</b>                                                                                                                                                                          |
|  <b>08.3</b>     | <b>Processed comminuted meat, poultry, and game products</b>                                                                                                                                                                                       |
|  <b>08.4</b>     | <b>Edible casings (e.g. sausage casings)</b>                                                                                                                                                                                                       |
|  <b>09.3</b>     | <b>Semi-preserved fish and fish products, including mollusks, crustaceans, and echinoderms</b>                                                                                                                                                     |

| Number                                                                                    | Food Category                                                                                                                                 |
|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
|  09.4     | Fully preserved, including canned or fermented fish and fish products, including mollusks, crustaceans, and echinoderms                       |
|  10.2.3   | Dried and/or heat coagulated egg products                                                                                                     |
|  10.3     | Preserved eggs, including alkaline, salted, and canned eggs                                                                                   |
|  10.4     | Egg-based desserts (e.g. custard)                                                                                                             |
|  11.6     | Table-top sweeteners, including those containing high-intensity sweeteners                                                                    |
|  12.2.2   | Seasonings and condiments                                                                                                                     |
|  12.3     | Vinegars                                                                                                                                      |
|  12.4     | Mustards                                                                                                                                      |
|  12.5     | Soups and broths                                                                                                                              |
|  12.6     | Sauces and like products                                                                                                                      |
|  12.7     | Salads (e.g. macaroni salad, potato salad) and sandwich spreads excluding cocoa- and nut-based spreads of food categories 04.2.2.5 and 05.1.3 |
|  12.8     | Yeast and like products                                                                                                                       |
|  12.9     | Soybean-based seasonings and condiments                                                                                                       |
|  12.10    | Protein products other than from soybeans                                                                                                     |
|  13.3     | Dietetic foods intended for special medical purposes (excluding products of food category 13.1)                                               |
|  13.4     | Dietetic formulae for slimming purposes and weight reduction                                                                                  |
|  13.5     | Dietetic foods (e.g. supplementary foods for dietary use) excluding products of food categories 13.1 - 13.4 and 13.6                          |
|  13.6     | Food supplements                                                                                                                              |
|  14.1.4   | Water-based flavoured drinks, including "sport," "energy," or "electrolyte" drinks and particulated drinks                                    |
|  14.2.1   | Beer and malt beverages                                                                                                                       |
|  14.2.2  | Cider and perry                                                                                                                               |
|  14.2.4 | Wines (other than grape)                                                                                                                      |
|  14.2.5 | Mead                                                                                                                                          |
|  14.2.6 | Distilled spirituous beverages containing more than 15% alcohol                                                                               |
|  14.2.7 | Aromatized alcoholic beverages (e.g. beer, wine and spirituous cooler-type beverages, low alcoholic refreshers)                               |
|  15.0   | Ready-to-eat savouries                                                                                                                        |
|  16.0   | Prepared foods                                                                                                                                |

Note: Unless otherwise specified, food additive provisions apply to the food category indicated (e.g. Dairy), as well as to all subcategories of that category (e.g. Cheese, Ripened Cheese, etc.).

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